

Large Language Model Agents in Repeated Public Goods Games: A Literature Review and Research Agenda

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Abstract

The public goods game (PGG) is the canonical model of the tension between individual self-interest and collective welfare, and its repeated form is the natural setting in which cooperation can be built, sustained, or eroded over time. As large language models (LLMs) are increasingly deployed as autonomous, mutually interacting agents, a fast-growing literature asks whether they cooperate, free-ride, reciprocate, punish, and form norms the way humans do. This review is organised around the intersection of these two threads—*LLM agents in repeated public goods games*—and is intended as an orienting map for new research in the area. We first consolidate the classical PGG background needed to read and benchmark agent studies: the formal model, the marginal per-capita return, the gap between the Nash equilibrium and the social optimum, the role of repetition, and the robust experimental and evolutionary regularities of human play (over-contribution with decay, conditional cooperation, the power of punishment, reputation, and institutional choice). We then survey, in depth, the emerging body of work on LLMs as strategic and economic agents, in repeated two-player social dilemmas, and—most directly—in public goods games and common-pool resource dilemmas, including studies of institutional sanctioning, emergent norms, and cooperation among populations of language models. Drawing the threads together, we find that current LLM-PGG research is concentrated on short horizons, a handful of models, and a narrow slice of the classical design space, leaving the long-run dynamics that define *repeated* cooperation largely unexplored. We close with a structured analysis of open problems and a concrete research agenda—spanning long-horizon dynamics, conditional cooperation, the reward–punishment asymmetry, the reasoning–cooperation trade-off, reputation, heterogeneous and mixed populations, threshold/collective-risk variants, and evolutionary analysis of LLM-authored strategies—meant to help identify where genuine novelty remains.

Keywords: public goods game; cooperation; repeated games; social dilemmas; large language models; multi-agent systems; free-riding; sanctioning institutions.

1 Introduction

Cooperation among self-interested actors is one of the central puzzles of the social and biological sciences (Nowak, 2006). Its sharpest formal expression is the *public goods game* (PGG): a group

of individuals each privately decide how much of an endowment to contribute to a common pool that is multiplied and shared equally, so that the group is best off when everyone contributes while each individual is privately best off contributing nothing (Samuelson, 1954; Olson, 1965; Ledyard, 1995). The PGG distils the “free-rider problem” and the “tragedy of the commons” (Hardin, 1968) into a few parameters, which is why it has become the workhorse of experimental economics, evolutionary biology, and political science for studying when and why cooperation succeeds or collapses.

Two features make the *repeated* public goods game especially important. First, almost all real collective-action problems—team production, taxation, environmental protection, open-source software, content moderation—are played out over many interactions rather than once. Second, repetition is precisely what opens the door to reciprocity, reputation, punishment, and norm formation: mechanisms that can turn a one-shot dilemma with a bleak toward defection into a setting where cooperation is an equilibrium (Van Segbroeck et al., 2012; Fehr and Gächter, 2000). Decades of human experiments and evolutionary models have mapped this repeated landscape in detail (Section 2).

A new class of decision-makers has now entered the picture. Large language models (LLMs) are increasingly deployed not as passive tools but as *autonomous agents* that plan, negotiate, and act in multi-agent systems (Park et al., 2023; Guo et al., 2024). When many such agents interact, their collective behaviour becomes a first-order question for both AI safety and the social sciences: do LLM agents cooperate for the common good, or do they free-ride; do they reciprocate and forgive; do they enforce norms; and does this behaviour match, exceed, or fall short of human cooperation? Because the PGG is the standard instrument for asking exactly these questions about humans, it is a natural and increasingly popular testbed for LLM agents as well (Guzman Piedrahita et al., 2025; Sreedhar et al., 2025; Piatti et al., 2024).

Purpose and scope. This article reviews the literature at the intersection of *LLM agents* and *repeated public goods games*, with two goals. The first is descriptive: to give a reader who is new to the area a coherent map of what has already been done—both the classical PGG results that any agent study must be read against, and the rapidly expanding set of LLM experiments in social dilemmas. The second is generative: to synthesise these findings into an explicit list of open problems and a research agenda, so that genuine novelty can be located rather than rediscovered. The review is therefore deliberately weighted toward the LLM literature (Sections 3–7), while keeping a compact but self-contained treatment of PGG theory and human evidence (Section 2) as the foundation and the benchmark.

Method. We draw on (i) the foundational and survey literature of public goods experiments and evolutionary game theory, and (ii) the 2022–2025 wave of papers that place LLMs in games, social dilemmas, and multi-agent simulations. For the latter we prioritise work that involves *repeated* interaction and/or the *public goods* / *commons* structure, while also covering closely related two-player and norm-formation studies that inform method and interpretation.

Roadmap. Section 2 establishes the PGG background: the formal model, the role of repetition, the main variants, and the robust regularities of human play. Section 3 reviews LLMs as strategic and economic agents. Section 4 surveys LLMs in repeated two-player games and social dilemmas. Section 5—the core of the review—examines LLMs in public goods and common-pool resource games, including sanctioning and emergent norms. Section 6 consolidates what is known and exposes the gaps. Section 7 proposes a concrete research agenda for LLMs in repeated PGGs. Section 8 concludes. Figure 1 gives a visual taxonomy of the reviewed literature, organised to mirror this structure.

2 Background: The Public Goods Game

This section provides the conceptual grounding needed to interpret and benchmark LLM studies. Readers familiar with the PGG may skim to Section 3; the stylized facts in Table 1 are the human baseline against which agent behaviour is repeatedly compared in later sections.

2.1 The linear public goods game

In the standard linear PGG, N players each receive an endowment e and simultaneously choose a contribution $c_i \in [0, e]$ to a common pool. The pool is multiplied by a factor α (with $1 < \alpha < N$) and divided equally. Player i 's payoff is

$$\pi_i = \underbrace{(e - c_i)}_{\text{kept}} + \frac{\alpha}{N} \sum_{j=1}^N c_j. \quad (1)$$

The ratio $\text{MPCR} = \alpha/N$ is the *marginal per-capita return*: the private return to each unit contributed. Because $\alpha < N$ implies $\text{MPCR} < 1$, every unit a player contributes yields them less than a unit kept, so contributing nothing is the dominant strategy and $c_i = 0$ for all i is the unique Nash equilibrium. Yet because $\alpha > 1$, total welfare is maximised when everyone contributes fully ($c_i = e$). The wedge between the individually rational outcome (universal free-riding) and the Pareto optimum (full contribution) is the social dilemma in its purest form (Ledyard, 1995; Camerer, 2003).

2.2 Why repetition matters

In a *one-shot* PGG, or in a *finitely* repeated PGG solved by backward induction, the theoretical prediction is unrelenting defection. The picture changes once interaction is genuinely repeated. In an *infinitely* (or indefinitely) repeated game, the folk theorem implies that cooperation can be sustained as a subgame-perfect equilibrium when players are patient enough, because future cooperation can be made contingent on present behaviour through strategies such as conditional cooperation and trigger strategies (Camerer, 2003). Repetition is thus not a technical detail but the very thing that makes reciprocity, reputation, punishment, and norm formation possible. It is also where human behaviour departs most strikingly from naive theory (Section 2.4), and—as this review emphasises—where the behaviour of LLM agents is least understood.

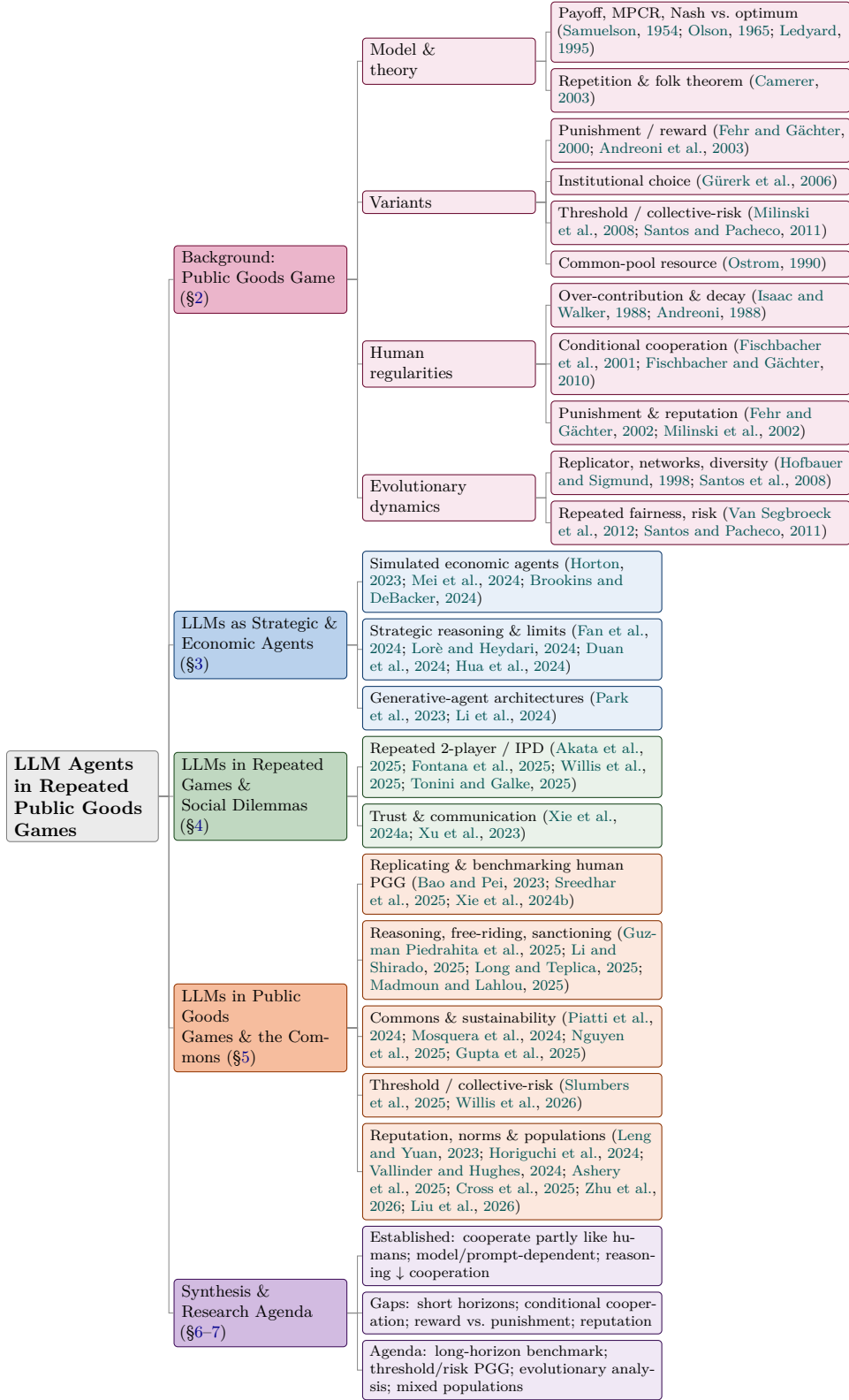


Figure 1: A taxonomy of the literature on LLM agents in repeated public goods games, mirroring the structure of this review. Branch colours denote the major parts: the PGG background (Section 2), LLMs as strategic and economic agents (Section 3), LLMs in repeated games and social dilemmas (Section 4), LLMs in public goods and commons games (Section 5, the core), and the synthesis and research agenda (Sections 6–7). Leaf nodes list representative references.

2.3 Variants relevant to agent studies

Several PGG variants recur in the agent literature and are worth naming:

- **PGG with punishment / reward.** After contributions are revealed, players may pay a cost to reduce (punish) or increase (reward) another’s payoff. Costly peer punishment dramatically raises cooperation in human experiments (Fehr and Gächter, 2000, 2002), and rewards and punishments have distinct effects (Andreoni et al., 2003).
- **Institutional / sanctioning choice.** Players choose between a sanction-free institution and one that permits punishment; over time people migrate to the sanctioning institution, which out-competes its rival (Gürrer et al., 2006). This design is directly adapted in recent LLM work (Guzman Piedrahita et al., 2025).
- **Threshold / collective-risk PGG.** A collective benefit (or the avoidance of a loss) is realised only if total contributions reach a threshold; otherwise players risk losing their endowment. This captures climate-change-type dilemmas (Milinski et al., 2008; Santos and Pacheco, 2011) and is discussed in Section 2.5.
- **Common-pool resource (CPR) / commons.** The “inverse” of the contribution PGG: players *extract* from a shared, regenerating resource, and over-extraction causes collapse (Ostrom, 1990). LLM versions of this dilemma are prominent (Piatti et al., 2024).
- **Structured populations.** Contributions interact through networks or are subject to evolutionary updating; heterogeneity and network structure strongly shape cooperation (Santos et al., 2008; Szabó and Fáth, 2007).

2.4 Robust regularities of human play

Five decades of laboratory work have produced a set of remarkably robust “stylized facts” (Table 1). People do *not* free-ride completely: initial contributions in linear PGGs typically average around 40–60% of the endowment (Ledyard, 1995; Zelmer, 2003). But under repetition without any cooperation-supporting mechanism, contributions *decay* toward (though rarely reaching) the free-riding prediction (Isaac and Walker, 1988; Andreoni, 1988). The decay is best explained not by pure confusion—though confusion contributes (Andreoni, 1995a; Houser and Kurzban, 2002)—but by the heterogeneity of types and *imperfect conditional cooperation*: roughly half of subjects raise their contribution when others do, but by slightly less than matching, so beliefs and contributions ratchet downward over time (Fischbacher et al., 2001; Fischbacher and Gächter, 2010; Croson, 2007). Framing, group size and the MPCR, communication, reputation, and—above all—the opportunity to punish free-riders are the main levers that move cooperation (Andreoni, 1995b; Isaac et al., 1994; Fehr and Gächter, 2000; Milinski et al., 2002; Chaudhuri, 2011). These regularities are the natural benchmark for any claim that LLM agents do or do not “behave like humans”.

2.5 Repeated-group and evolutionary dynamics

Beyond the laboratory, evolutionary game theory studies how cooperation fares when strategies are updated over time within a population, typically through the replicator dynamics $\dot{x} = x(1-x)(f_C - f_D)$, where x is the fraction of cooperators and f_C, f_D their average payoffs

Table 1: Robust regularities of human behaviour in (repeated) public goods games. These constitute the empirical baseline against which LLM agents are compared in Sections 5–6.

Regularity	Summary and key references
Partial cooperation, not full free-riding	Initial contributions average $\sim 40\text{--}60\%$ of endowment, far above the Nash prediction of zero (Ledyard, 1995; Zelmer, 2003).
Decay under repetition	Without supporting mechanisms, contributions decline toward zero over rounds (Isaac and Walker, 1988; Andreoni, 1988).
Conditional cooperation	$\sim 50\%$ of people are conditional cooperators who match others imperfectly; $\sim 30\%$ are free-riders (Fischbacher et al., 2001; Fischbacher and Gächter, 2010).
Kindness <i>and</i> confusion	Cooperation reflects genuine other-regard plus decision error in roughly equal measure (Andreoni, 1995a; Houser and Kurzban, 2002).
Framing effects	Positive (“warm-glow”) vs. negative framing changes cooperation for identical payoffs (Andreoni, 1995b).
MPCR and group size	Higher MPCR raises contributions; group-size effects are largely an MPCR artefact (Isaac and Walker, 1988; Isaac et al., 1994).
Punishment sustains cooperation	Costly peer punishment sharply raises and stabilises contributions, even when individually costly (Fehr and Gächter, 2000, 2002).
Reputation and institutions	Indirect reciprocity and a choice of sanctioning institutions both rescue cooperation (Milinski et al., 2002; Gürer et al., 2006; Rockenbach and Milinski, 2006).

(Hofbauer and Sigmund, 1998; Sigmund, 2010). In well-mixed populations cooperation tends to vanish, but a battery of mechanisms can reverse this: network reciprocity and social diversity (Santos et al., 2008; Szabó and Fáth, 2007; Perc et al., 2017), optional participation (Hauert et al., 2002), the emergence of costly punishment (Hauert et al., 2007; Boyd et al., 2003), institutions learned through social imitation (Sigmund et al., 2010), and multilevel selection (Traulsen and Nowak, 2006); Nowak (2006) organises these into five canonical rules.

Two strands are especially relevant to *repeated* PGGs and to the LLM agenda. First, Van Segbroeck et al. (2012) model the *repeated* N -person game directly: agents are “reciprocators” who contribute only if at least M others contributed in the previous round, and evolution selects for an intermediate, moderate notion of fairness M^* , producing cyclic alternations between cooperation and defection rather than a static equilibrium. Second, the *collective-risk* dilemma combines repetition with a threshold and the risk of catastrophic loss: in Milinski et al.’s (2008) climate experiment, groups reached the target only when the perceived risk of loss was high, and Santos and Pacheco (2011) show analytically that a high risk of collective failure—together with small group size and stringent thresholds—is what turns cooperation from a losing strategy into a viable one. These results give concrete, quantitative hypotheses (the role of risk, thresholds, group size, and reciprocation rules) that LLM studies have barely begun to test.

3 LLMs as Strategic and Economic Agents

Before turning to multi-agent public goods settings, it is useful to summarise what is known about individual LLM behaviour in economic and strategic tasks, because these capabilities and biases propagate into the PGG.

LLMs as simulated economic agents. Horton (2023) argued that an LLM endowed with a persona can serve as “*homo silicus*”—a simulated economic agent whose responses reproduce classic behavioural results, suggesting LLMs can act as in-silico subjects for economics. Mei et al. (2024) formalised this with a behavioural Turing test across a battery of personality measures and games (dictator, trust, public goods, prisoner’s dilemma), finding that GPT-4 generally behaves within the human distribution and is often *more* cooperative and altruistic than the median person. Brookins and DeBacker (2024) similarly found GPT to display stronger fairness and higher cooperation than typical human benchmarks in dictator and prisoner’s dilemma games. The recurring theme is partial human-likeness with a prosocial tilt—and substantial sensitivity to how the situation is framed.

Strategic reasoning: capability and limits. A parallel line evaluates whether LLMs reason strategically in the game-theoretic sense. Results are mixed. Fan et al. (2024) report that even GPT-4 deviates systematically from rational benchmarks in dictator, rock–paper–scissors, and ring-network games. Lorè and Heydari (2024) disentangle the effects of formal game structure from contextual framing across the prisoner’s dilemma, stag hunt, and snowdrift, showing that LLMs are swayed by surface framing more than humans. Duan et al. (2024) introduce GTBENCH, a suite of game-theoretic tasks, and find LLMs strong on probabilistic, incomplete-information games but weak on deterministic, complete-information ones. Hua et al. (2024) show that deviations from equilibrium grow with game complexity and propose explicit “game-theoretic workflows” to improve equilibrium computation, while Gandhi et al. (2023) demonstrate that structured chain-of-thought prompting (tracking states, values, and beliefs) markedly improves strategic play. The implication for PGGs is twofold: LLM cooperation is real but is mediated by prompting and framing, and “better reasoning” does not automatically translate into better strategic—or more cooperative—behaviour.

Generative-agent architectures. Methodologically, much multi-agent PGG work builds on the generative-agent paradigm of Park et al. (2023), in which agents are equipped with memory, reflection, and planning modules and exhibit believable emergent social behaviour in a sandbox. Li et al. (2024) extend this to macroeconomic simulation with perception/memory/reflection agents. These architectures supply the machinery—persistent memory, identities, and natural-language interaction—that distinguishes a multi-agent LLM PGG from a sequence of isolated prompts, and that will be central to studying *repeated* cooperation.

4 LLMs in Repeated Games and Social Dilemmas

The closest precedent to repeated PGGs is the study of LLMs in repeated two-player social dilemmas, where the dynamics of reciprocity and forgiveness can be observed cleanly.

Repeated two-player games. Akata et al. (2025) systematically played families of repeated 2×2 games with LLMs and found a characteristic profile: LLMs coordinate well but are *unforgiving* in the iterated prisoner’s dilemma, defecting persistently after a single defection by the opponent—behaviour that is individually defensible but corrosive to long-run cooperation.

Fontana et al. (2025), by contrast, report that several models are “nicer than humans” in the iterated prisoner’s dilemma, though newer models can be more exploitative, underscoring that conclusions are model-dependent. Phelps and Russell (2023) use experimental-economics prompts to probe emergent goal-like dispositions and show that persona and framing strongly steer cooperation. Willis et al. (2025) take an evolutionary view: they prompt LLMs to *author* full iterated-PD strategies and then analyse those strategies with evolutionary game theory, revealing systematic, model-specific biases toward aggression or cooperation. Tonini and Galke (2025) show that pairing within-team repetition with between-team competition yields “super-additive” cooperation, raising even one-shot cooperation rates—an LLM echo of multilevel selection.

Trust, communication, and other multi-agent games. Xie et al. (2024a) find, using behavioural trust games, that LLM agents exhibit trust behaviour aligning closely with humans for GPT-4. Communication-rich games such as Werewolf reveal emergent strategic behaviours (trust, bluffing, leadership) from frozen models (Xu et al., 2023). Together these studies establish that LLM agents can reciprocate, trust, and communicate strategically over repeated interactions—but also that they carry strong, idiosyncratic, and prompt-sensitive dispositions that must be controlled for when moving to the N -player public goods setting.

5 LLMs in Public Goods Games and the Commons

We now reach the core of the review: studies that place LLM agents directly in public goods or common-pool resource dilemmas. Table 2 summarises the key studies along the design dimensions that matter most for repeated cooperation.

5.1 Replicating human PGG behaviour

The earliest direct placements of an LLM in a PGG simply elicited contributions: Bao and Pei (2023) had GPT play a voluntary-contribution game and linked its contribution level to cognitive uncertainty. Behavioural-economics benchmark batteries now routinely include the PGG among a suite of canonical games (Mei et al., 2024; Xie et al., 2024b; Leng and Yuan, 2023), generally finding that capable LLMs contribute at or above human levels, but with pronounced variation across model families.

Sreedhar et al. (2025) ask whether multi-agent LLM systems can reproduce human prosocial behaviour in the PGG. Using a generative-agent architecture with three or four player agents, they replicate three classic human treatments—name framing (“Teamwork” vs. “Taxation”), transparency of contributions, and varying endowments—and find that LLM agents reproduce the *direction* of each human effect (e.g., priming toward teamwork raises contributions; transparency raises contributions; large endowments lower the contributed fraction), although they tend to *overestimate the magnitude*. They further show that effects can transfer from non-PGG experiments and that, with added communication channels, agents display “unbounded” real-world behaviours such as collaborating and even cheating. The study is important evidence that LLMs encode a usable model of human cooperative tendencies, and that this model is steerable

by the same levers (framing, transparency, endowment) that move humans. Its limits—mostly one-shot or short (five-round) games, a single primary model, no sanctioning—also mark where repeated-PGG work must go.

5.2 Reasoning, free-riding, and institutional sanctioning

The work closest to a genuine *repeated PGG with institutions* is [Guzman Piedrahita et al. \(2025\)](#). Adapting the institutional-choice design of [Gürerk et al. \(2006\)](#), they run a 15-round PGG with seven LLM agents who each round choose between a *sanctioning institution* (SI), in which members may pay to reward (+1 at cost 1) or punish (−3 at cost 1) one another, and a *sanction-free institution* (SFI); contributions are multiplied by 1.6 and shared within the chosen institution. Their findings are striking and directly relevant:

- **Four behavioural archetypes** emerge across models—*increasingly cooperative*, *increasingly defecting*, *no change*, and *unstable (oscillating)*—closely mirroring the heterogeneity seen in human and evolutionary studies.
- **Reasoning models cooperate less.** Traditional models (e.g., GPT-4o, DeepSeek-V3, Llama-3.3-70B) sustain high cooperation comparable to humans, whereas reasoning-optimised models (the o1/o3 family) frequently collapse to free-riding—“corrupted by reasoning”. This is a pointed counterexample to the assumption that stronger reasoning yields better collective outcomes.
- **Reward, not punishment.** Although capable LLMs replicate human cooperative *outcomes*, they do so through a different *mechanism*: humans rely heavily on punishment (~ 1.66 punish/reward ratio), whereas all LLMs strongly prefer rewards, raising doubts about whether they grasp deterrence and the stability of the cooperation they achieve.

This study is the clearest demonstration to date that the *repeated*, institution-rich PGG is a revealing probe of LLM cooperation, and that architecture (reasoning vs. not) is a first-order determinant of outcomes.

The headline result—that reasoning optimisation can *erode* cooperation—has since been corroborated and extended. [Li and Shirado \(2025\)](#) find, across the PGG and five other cooperation games, that eliciting chain-of-thought “calculated” reasoning systematically lowers giving and norm enforcement relative to more “spontaneous” responses, and [Backmann et al. \(2025\)](#) (Moral-Sim) show that no frontier model stays cooperative once payoffs conflict with stated moral norms in repeated prisoner’s dilemma and PGG play. [Long and Teplica \(2025\)](#) add a twist: in an iterated PGG, telling a model that its co-player is “another AI” versus “itself” measurably shifts contributions. A complementary line uses the repeated PGG constructively: [Madmoun and Lahlou \(2025\)](#) show that cheap-talk communication restores cooperation in an iterated PGG with punishment (while a rationality-maximising training curriculum suppresses it), [Warnakulasuriya et al. \(2025\)](#) replicate punishment-driven norm emergence in an N -player sanctioning dilemma, and [Liang et al. \(2025\)](#) redesign a sequential PGG as a mechanism that makes contribution the unique equilibrium for multi-LLM ensembles. The consistent thread is that, as for humans, LLM cooperation in the repeated PGG is fragile to “rational” framing and strongly helped by

communication and sanctions.

5.3 Common-pool resources and sustainability

Piatti et al. (2024) introduce GOVSIM, a common-pool resource simulation (fishery/pasture/pollution variants) in which LLM agents repeatedly harvest a regenerating resource. Most models over-harvest and drive the resource to collapse; only the strongest models, and only when allowed to *communicate* and to engage in “universalisation” moral reasoning (“what if everyone did this?”), achieve sustainable cooperation. GOVSIM is the commons (or “inverse”) counterpart to the contribution PGG and demonstrates two themes that recur throughout this review: communication is a powerful lever for LLM cooperation, and sustainability over long horizons is fragile and model-dependent.

The commons side has since become comparatively well populated. Mosquera et al. (2024) were among the first to place LLM-augmented agents in DeepMind’s MELTING POT “Commons Harvest” substrate, and Chacón-Chamorro et al. (2025) use the same family of environments to compare the *cooperative resilience* of humans and LLMs under stochastic disruptions. Building on GOVSIM, Nguyen et al. (2025) equip agents with an explicit “consideration of future consequences” to sustain shared resources, while Gupta et al. (2025) embed Ostrom-style social learning and norm-based punishment so that conservation norms emerge endogenously without reward shaping. Across these studies the message echoes Piatti et al. (2024): collapse is the default outcome, and communication, foresight, and norms are what avert it.

5.4 Threshold and collective-risk dilemmas

The threshold / collective-risk variant—contribute enough to avert a catastrophic loss, or risk losing everything (Section 2.5)—is the LLM niche most directly connected to the climate-style dilemmas of Milinski et al. (2008) and Santos and Pacheco (2011), and it remains almost empty. The closest work is Slumbers et al. (2025), who have LLM agents play *Port of Mars*, a multi-round collective-risk social dilemma with a catastrophe threshold; however, it is framed as a test of whether LLMs can substitute for human experimental subjects rather than as a study of threshold-cooperation dynamics, and it uses a single model. Willis et al. (2026) include a collective-risk dilemma among three population-scale games, but in a paradigm where models *author* strategies that are then executed automatically, rather than deliberating round by round. A dedicated, multi-model study of LLM agents *playing* a repeated collective-risk PGG—probing risk sensitivity, threshold behaviour, group size, and framing—does not yet appear to exist, a gap we return to in Section 7.

5.5 Reputation, norms, and populations of agents

A further cluster studies cooperation and norms among *populations* of LLM agents. Leng and Yuan (2023) run canonical behavioural-economics games (including the PGG) and show that LLM reasoning invoking fairness or altruism predicts prosocial choices. Horiguchi et al. (2024) find that agents conversing in natural language can spontaneously develop social norms and even *metanorms* (punishing those who fail to punish), echoing the second-order free-rider

problem central to human cooperation theory. Vallinder and Hughes (2024) study indirect reciprocity across *generations* of agents in an iterated Donor Game and find that cooperation emerges for some model families (e.g., Claude) but not others, and is shaped by the availability of punishment and by cultural transmission. Ashery et al. (2025) show that decentralised LLM populations spontaneously converge on shared conventions, with tipping points driven by committed minorities. These studies collectively show that the ingredients of repeated-PGG cooperation—reciprocity, reputation, punishment, norms—can all emerge among LLM agents, but unevenly and with strong model and design dependence.

Two further threads sharpen this picture. On *reputation*, Cross et al. (2025) show that generative LLM agents reproduce the empirical boost that gossip and ostracism give to PGG contributions, and Zhu et al. (2026) build a decentralised gossip network in which self-interested agents form reputations and ostracise defectors—directly importing indirect reciprocity into the LLM PGG. On *scale and horizon*, Willis et al. (2026) push social-dilemma simulations to hundreds of agents (finding that newer models trend toward *worse* collective equilibria), Liu et al. (2026) run social dilemmas including PGG-type games for 500 rounds and report a “memory curse” in which longer recall *erodes* cooperation, and Tewolde et al. (2026) benchmark cooperation-sustaining mechanisms (repetition, reputation, mediation, contracts) across prisoner’s dilemma and public-goods settings.

6 Synthesis: What Is Established, and What Is Missing

6.1 What the literature has established

Reading the agent studies against the human baseline of Section 2, several conclusions are now reasonably well supported:

1. **LLMs can cooperate, and partly like humans.** Across PGGs and related games, capable LLMs contribute substantially and reproduce the *direction* of major human effects (framing, transparency, endowment, benefits of communication) (Sreedhar et al., 2025; Mei et al., 2024; Piatti et al., 2024).
2. **Behaviour is strongly model- and prompt-dependent.** Cooperation varies sharply across model families and is sensitive to framing and persona (Fontana et al., 2025; Lorè and Heydari, 2024; Vallinder and Hughes, 2024); conclusions rarely generalise across all models.
3. **Reasoning optimisation can reduce cooperation.** The reasoning-tuned o1/o3 family free-rides more than standard models in the repeated institutional PGG (Guzman Piedrahita et al., 2025), and “calculated” chain-of-thought lowers giving across the PGG and related games (Li and Shirado, 2025; Backmann et al., 2025)—complicating the assumption that capability implies prosociality.
4. **Mechanism, not just outcome, differs from humans.** LLMs lean on rewards rather than punishment (Guzman Piedrahita et al., 2025) and can be unforgiving after a single defection (Akata et al., 2025)—outcomes can match humans while the underlying strategy does not.

5. **The ingredients of repeated cooperation can emerge.** Reciprocity, reputation, punishment, norms, and even metanorms appear among LLM agents (Horiguchi et al., 2024; Vallinder and Hughes, 2024; Ashery et al., 2025), but unevenly.

6.2 Methodological gaps

The field is young, and its methods are correspondingly narrow:

- **Short horizons.** Most “repeated” LLM PGGs run only a handful to a few tens of rounds (Table 2); only isolated recent work reaches hundreds of rounds (Liu et al., 2026). The long-run phenomena that *define* repeated cooperation—steady-state contribution decay, endgame effects, the slow build-up of reputation, cyclic cooperation/defection (Van Segbroeck et al., 2012)—remain largely untested.
- **Few models, weak statistics.** Studies often use one or two proprietary models, sometimes with single runs per condition, limiting statistical power and reproducibility.
- **Narrow design slice.** Communication, persistent identity, reputation, threshold/risk structures, and large groups are each present in only a subset of studies, and rarely in combination.
- **No standard benchmark.** There is no agreed-upon repeated-PGG protocol, set of metrics, or human-baseline comparison, making cross-study synthesis difficult.

6.3 Conceptual gaps mapped to the classical literature

More importantly, mapping agent studies onto the human/evolutionary PGG literature exposes specific, high-value questions that remain open:

- **Conditional cooperation.** Are LLMs conditional cooperators in the precise sense of Fischbacher et al. (2001)—and do they exhibit the same imperfect, self-serving matching that drives human decay? A strategy-method analog for LLMs is largely missing.
- **Contribution decay.** Do LLM contributions decay over long horizons as human contributions do (Isaac and Walker, 1988; Fischbacher and Gächter, 2010), and is any decay driven by belief updating or by context-window/memory effects—as the “memory curse” of Liu et al. (2026) begins to suggest?
- **Reward–punishment asymmetry.** Why do LLMs prefer reward to punishment (Guzman Piedrahita et al., 2025), is it robust across designs, and does it create a second-order free-rider problem in enforcement?
- **The reasoning–cooperation trade-off.** What mechanism makes reasoning models free-ride—chain-of-thought toward “rational” defection, RLHF objectives, or framing? Can alignment or prompting restore cooperation without harming capability?
- **Reputation and indirect reciprocity.** Human cooperation is rescued by reputation (Milinski et al., 2002; Rockenbach and Milinski, 2006), yet leading LLM PGGs *anonymise* agents each round (Guzman Piedrahita et al., 2025). First steps re-introduce gossip and reputation (Cross et al., 2025; Zhu et al., 2026), but persistent identities and indirect reciprocity in the institutional PGG remain largely open.

- **Heterogeneous and mixed populations.** Almost all studies use homogeneous model populations. Mixed-model societies, and LLM–human mixed groups, are barely explored despite their practical importance.
- **Threshold/collective-risk PGGs.** The risk-driven cooperation of Milinski et al. (2008) and Santos and Pacheco (2011) has only just begun to be probed with LLMs—through a single-model collective-risk simulation (Slumbers et al., 2025) and a strategy-authoring population study (Willis et al., 2026). Whether LLM agents *deliberating* round by round are risk-sensitive in the way that rescues human cooperation, and how this varies across models and framings, remains essentially unknown.
- **Evolutionary/population analysis.** Willis et al. (2025) analyse LLM-authored *IPD* strategies with evolutionary game theory, and Willis et al. (2026) extend the strategy-authoring paradigm to PGG, collective-risk, and commons games at population scale; a complementary analysis of LLM agents *playing* the N -person PGG turn by turn (their ESS, replicator dynamics, invasion by free-riders) is still open.

7 A Research Agenda for LLMs in Repeated Public Goods Games

The gaps above suggest a coherent programme. We organise it as a design space plus a set of concrete, novel directions.

A design space for repeated-PGG agent experiments. A study can be positioned along independent axes, most of which are currently under-sampled: (i) *horizon* (few rounds vs. hundreds; known vs. unknown endpoint, to expose endgame effects); (ii) *enforcement* (none / peer punishment / reward / pool punishment / institutional choice (Gürerk et al., 2006; Sigmund et al., 2010)); (iii) *information* (anonymous vs. persistent identities and reputation); (iv) *communication* (none vs. free-form negotiation, shown to be pivotal in (Piatti et al., 2024)); (v) *population* (single model, mixed models, LLM–human mixes); (vi) *payoff structure* (linear vs. threshold/collective-risk vs. commons); and (vii) *memory* (context window, summarisation, external memory).

Promising and largely novel directions.

1. **Long-horizon dynamics and benchmarking against human stylized facts.** Run PGGs over many rounds and test directly for decay, conditional cooperation, and endgame effects against Table 1; only isolated studies yet reach this regime (Liu et al., 2026). A standardised, open repeated-PGG benchmark (fixed protocol, metrics, human baselines)—in the spirit of emerging mechanism benchmarks (Tewolde et al., 2026)—would itself be a substantial contribution.
2. **A strategy-method characterisation of LLM conditional cooperation.** Adapt Fischbacher et al. (2001) to elicit each model’s contribution as a function of others’ average contribution, classifying models as conditional cooperators, free-riders, or other types.

3. **Mechanistic study of the reasoning–cooperation trade-off.** Causally isolate why reasoning models defect (Guzman Piedrahita et al., 2025) by varying chain-of-thought, decoding, persona, and alignment, and test interventions that restore cooperation.
4. **Reputation and indirect reciprocity with persistent identities.** Building on first gossip-based studies (Cross et al., 2025; Zhu et al., 2026), re-introduce persistent identity and reputation into the institutional PGG and test whether LLM cooperation responds as human cooperation does (Milinski et al., 2002; Rockenbach and Milinski, 2006).
5. **The reward–punishment asymmetry and second-order free-riding.** Systematically probe why LLMs avoid punishment, whether they sustain enforcement, and whether metanorms (Horiguchi et al., 2024) stabilise it.
6. **Threshold and collective-risk PGGs with LLMs.** Port the climate-style dilemma (Milinski et al., 2008; Santos and Pacheco, 2011) to LLM agents and measure risk sensitivity, threshold behaviour, and the effect of group size. Existing work either uses a single model or has models author strategies (Slumbers et al., 2025; Willis et al., 2026); a systematic, multi-model study of agents *deliberating* round by round would occupy a still-open, high-impact niche.
7. **Evolutionary analysis of LLM PGG strategies.** Where Willis et al. (2025) and Willis et al. (2026) have models *author* strategies, complement this by extracting the effective strategies of agents that *play* the N -person PGG turn by turn, then study their evolutionary stability and invasibility under replicator dynamics.
8. **Heterogeneous and mixed-population societies.** Compose groups of different model families, and of LLMs with humans, to study cooperation, exploitation, and convention formation (Ashery et al., 2025) in realistic mixed settings.
9. **Cooperation as an AI-safety property.** Frame multi-agent cooperation in repeated PGGs as an alignment target, connecting the empirical findings to the governance of deployed multi-agent systems (Guo et al., 2024; Sun et al., 2025).

Taken together, these directions share a common logic: the human and evolutionary PGG literature provides a rich, quantitative set of hypotheses about *repeated* cooperation, while current LLM studies have tested only a small corner of it. The clearest novelty therefore lies in (a) genuinely long-horizon experiments, (b) designs that re-introduce the mechanisms humans rely on (punishment, reputation, risk), and (c) bringing evolutionary and benchmarking rigour to LLM-authored strategies.

8 Conclusion

The public goods game has been, for half a century, the sharpest lens on the conflict between private incentive and collective good, and its repeated form is where the mechanisms that rescue cooperation—reciprocity, reputation, punishment, norms, and institutions—actually operate. As large language models become interacting agents, the same lens is now being turned on them. The emerging evidence is that capable LLMs cooperate substantially and reproduce many human regularities, that their behaviour is highly model- and prompt-dependent, that reasoning optimisation can paradoxically erode cooperation, and that they often achieve cooperative

outcomes through non-human mechanisms (reward over punishment, unforgiving reciprocity). Yet the studies establishing these points have so far explored only a narrow slice of the classical design space, and—crucially for anyone interested in *repeated* cooperation—the long-horizon dynamics that define the repeated PGG remain largely uncharted. That gap, mapped in detail above, is also the opportunity: a disciplined programme that takes the repeated PGG seriously, at realistic horizons, with the mechanisms humans rely on and the rigour of evolutionary game theory, is well positioned to produce genuinely novel results on how—and whether—populations of language models can sustain the common good.

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Table 2: Representative studies of LLM agents in public goods, commons, and closely related social dilemmas, along the design dimensions most relevant to *repeated* cooperation. “Rounds” indicates the degree of repetition; “Sanction” indicates whether punishment/reward or institutional choice is available; “Comm.” indicates free-form natural-language communication.

Study	Game	Rounds	Sanc.	Comm.	Key finding
Sreedhar et al. (2025)	PGG (contribution)	1–5	No	Partly	LLMs reproduce the direction of human framing/transparency/endowment effects, but overshoot the magnitude.
Guzman Piedrahita et al. (2025)	PGG + institutional choice	15	Yes	No	Reasoning models free-ride; traditional models cooperate; all LLMs prefer reward over punishment.
Li and Shirado (2025)	PGG (+5 coop. games)	Many	Yes	No	Chain-of-thought “calculated” reasoning lowers giving and norm enforcement.
Piatti et al. (2024)	Common-pool resource	Many	No	Yes	Most models over-harvest and collapse; communication + universalisation enable sustainability.
Slumbers et al. (2025)	Collective-risk (Port of Mars)	9	No	Yes	LLMs play a catastrophe-threshold dilemma; tested as human-subject surrogates.
Nguyen et al. (2025)	Common-pool resource	Many	No	Yes	“Consideration of future consequences” sustains the shared resource.
Leng and Yuan (2023)	PGG, dictator, PD	1	No	No	Reasoning that cites fairness/altruism predicts prosocial LLM actions.
Horiguchi et al. (2024)	Norms game	Many	Yes	Yes	Agents develop norms and metanorms (punishing non-punishers) in natural language.
Vallinder and Hughes (2024)	Donor game (indirect recip.)	Many (gen.)	Yes	No	Cooperation emerges for some model families; shaped by punishment and cultural transmission.
Akata et al. (2025)	Repeated 2×2 (incl. IPD)	Many	No	No	LLMs coordinate well but are unforgiving in the IPD.
Willis et al. (2025)	Iterated PD (evolutionary)	Many	No	No	LLM-authored strategies show model-specific aggression/cooperation biases.
Tonini and Galke (2025)	IPD + group competition	Many	No	No	Within-team repetition + between-team competition yields super-additive cooperation.
Cross et al. (2025)	PGG + gossip/ostracism	Many	Yes	Yes	Generative agents reproduce reputation and punishment effects on contributions.
Liu et al. (2026)	Social dilemmas (incl. PGG)	500	No	No	“Memory curse”: longer recall erodes cooperative intent over long horizons.